4.1. Antidiabetic Drugs

Several drugs for type 2 diabetes also act on pathways that drive inflammation in RA, such as AMP-activated protein kinase (AMPK) and NF-κB (Section 3). In RA, insulin resistance also correlates weakly with disease activity ($\rho = 0.21$) and more strongly with inflammatory cytokines such as interleukin-6 ($\rho = 0.63$) [1]. This section first covers metformin and pioglitazone, followed by glucagon-like peptide-1 receptor agonists (GLP-1RAs) and sodium–glucose cotransporter-2 (SGLT2) inhibitors.

4.1.1. Metformin

Metformin is a common oral treatment for type 2 diabetes, but how it acts at the cellular level was long unclear [2]. The drug appears to act mainly by activating AMPK, which in hepatocytes lowers acetyl-CoA carboxylase activity, increases fatty acid oxidation and suppresses lipogenic enzyme expression [2]. Activated AMPK phosphorylates raptor, a step needed to inhibit mammalian target of rapamycin (mTOR) complex 1 (mTORC1) under energy stress [3]. Both mTOR complexes, mTORC1 and mTORC2, promote T helper 17 (Th17) cell differentiation [4].

Two animal studies showed that metformin reduced arthritis and protected the joint. In the collagen-induced arthritis (CIA) model (Table 1), immunized mice were given 5 mg/mouse of metformin orally three times per week for 9 weeks, starting 7 days after the first immunization. This treatment significantly reduced arthritis severity and shifted the Th17 to regulatory T (Treg) cell ratio, with fewer Th17 cells and more Treg cells. The numbers of T cells positive for phosphorylated signal transducers and activators of transcription 3 and 5 (STAT3 and STAT5) fell and rose, respectively, and osteoclast activity was reduced [5]. Similarly, in rats with CIA, intraperitoneal (i.p.) metformin (100 mg/kg every 3 days for 3 weeks) protected the knee joints on micro-computed tomography (micro-CT) and on histopathology. Micro-CT showed preserved trabecular bone, histology showed less osteoclast formation and synovitis, and metformin also lowered markers of systemic inflammation, chondrocyte apoptosis and cartilage matrix degradation [6]. After body-surface-area conversion for a 60-kg adult [7], the oral dose of 5 mg/mouse [5] equals approximately 810–1,220 mg per dose. This range assumes a 20–30-g mouse,

because body weight was not reported. The i.p. dose of 100 mg/kg in rats [6] equals approximately 970 mg per dose. These per-dose values lie within the usual clinical range of 500–2,550 mg/day [8]. Averaged over the dosing intervals of three times weekly and every 3 days, however, they correspond to approximately 350–520 mg/day and 320 mg/day, respectively, at or below the lower limit of that range [7].

Human joint tissue data also suggest a link between AMPK and RA activity. RA synovium expressed more AMPK-α1 than osteoarthritis synovium; in RA serum, phosphorylated AMPK-α1 correlated weakly with the 28-joint Disease Activity Score (DAS28; $r = 0.27$), and total AMPK-α1 correlated moderately with tumor necrosis factor-α (TNF-α; $r = 0.46$) [9]. In cultured RA fibroblast-like synoviocytes (FLS), metformin inhibited proliferation and migration at millimolar concentrations (5–10 mM) [9]. It also lowered the levels of TNF-α, interleukin (IL)-1β and IL-6 in these cells [9].

Three clinical studies suggest a benefit of metformin as an add-on to conventional synthetic (cs) disease-modifying anti-rheumatic drugs

(DMARDs). The larger of the two randomized trials, a double-blind study from Egypt, assigned 120 patients with active RA to metformin 1,000 mg or placebo daily, each added to methotrexate (MTX) 7.5 mg/week, over 12 weeks [10]. At 12 weeks, 80.8% of the metformin group reached an American College of Rheumatology 20% improvement (ACR20) response, compared with 54.7% of the placebo group (P = 0.001), without serious adverse effects in either group [10]. A randomized, single-blind trial enrolled 66 Egyptian patients with moderately or highly active RA, of whom 33 received metformin 850 mg twice daily in addition to csDMARDs and 33 received placebo. After 6 months, C-reactive protein (CRP) levels and DAS28-CRP scores were lower than with placebo among the 60 patients who completed the trial, and three metformin-treated patients (9%) withdrew because of gastrointestinal side effects [11]. Because these withdrawals were excluded from the analysis, the benefit may be overestimated. In a non-randomized prospective cohort study, 124 patients received low-dose metformin (250 mg twice daily) in addition to traditional csDMARD therapy for 90 days and were compared with 98 patients receiving

csDMARDs only. Metformin use was associated with a faster fall in IL-6, TNF-α, high mobility group box 1 (HMGB1) and CRP than in the csDMARD-only group [12]. A meta-analysis of four randomized trials with 350 patients found that metformin lowered CRP after 3 months (mean difference −5.99 mg/L, 95% confidence interval [CI] from −7.46 to −4.52) [13]. Over the same period, DAS28 also fell in the three trials that reported it (mean difference −0.80, 95% CI from −0.87 to −0.73) [13]. This interval appears implausibly narrow. The three trials were small, and DAS28 varied by about 1.5 units between patients [10]; the pooled estimate should therefore be read with caution. However, each of the four trials was conducted in Egypt or India, and two carried a high risk of bias because they were single-blind and did not report trial registration [13]. Therefore, metformin may be a well-tolerated add-on to csDMARDs for lowering RA disease activity in short trials, but larger and longer double-blind randomized controlled trials (RCTs) that also check joint damage on X-ray are needed.

Data on RA prevention are conflicting. In a large UK cohort of people recently diagnosed with type 2 diabetes, metformin use showed no clear

association with RA risk compared with other oral diabetes drugs (adjusted hazard ratio [HR] 1.00, 95% CI from 0.82 to 1.22) [14]. A Mendelian randomization study suggested that metformin may protect against RA (odds ratio [OR] 0.023, 95% CI from 0.0016 to 0.34) [15]. This effect size is implausibly large for a drug effect and should be interpreted with caution. Genetic variants linked to metformin use may also reflect the genetic liability to diabetes itself. In an Israeli cohort of 113,749 people starting metformin, adherence was associated with a lower RA risk, which was lowest when 40–59% of days were covered rather than under 20% (adjusted HR 0.62, 95% CI from 0.45 to 0.84) [8]. Because the risk was lowest at intermediate rather than the highest adherence, the association did not rise steadily with adherence, which leaves healthy-adherer bias and chance as plausible explanations. In analyses stratified by sex, this association was limited to women [8].

4.2. Lipid-Lowering Drugs

Patients with RA have up to twice the risk of atherosclerotic cardiovascular disease seen in the general population, a level similar to that in diabetes [16], which makes lipid-lowering drugs a logical choice to test. The two classes examined

here, statins and fenofibrate, also affect immune cells and may benefit both the joints and the heart (Sections 4.2.1 and 4.2.2).

4.2.1. Statins

Statins act mainly as competitive inhibitors of 3-hydroxy-3-methylglutaryl-CoA reductase (HMG-CoA reductase), binding the enzyme at nanomolar concentrations and displacing its natural substrate, which blocks de novo cholesterol synthesis [17]. Statins also have effects beyond lipid lowering, because fewer isoprenoid intermediates leave small guanosine triphosphatases (GTPases) such as Ras, Rho, Rac and Cdc42 less prenylated and therefore less active [17]. They also block interferon-γ (IFN-γ)-induced major histocompatibility complex (MHC) class II expression and thus MHC-dependent T-cell activation [18], and T cells form the largest lymphocyte population in rheumatoid synovium [19].

Animal data on statins are conflicting. In a preclinical study, mice with serum-induced arthritis were given atorvastatin, pravastatin or simvastatin intravenously (0.2 mg/kg on days 3, 5 and 7), and atorvastatin and pravastatin lessened

clinical arthritis and preserved joint structure on paw histology, but simvastatin had no significant effect [20]. This regimen corresponds to approximately 1 mg per dose in a 60-kg adult after body-surface-area conversion [7], far below the 40 mg/day used in RA trials [21, 22], although intravenous and oral doses are not directly comparable.

An earlier CIA study with simvastatin was favorable. In mice given daily i.p. simvastatin at 10, 20 or 40 mg/kg between days 21 and 40, arthritis was suppressed in a dose-dependent manner [23], and a 14-day course of 40 mg/kg, started one day after clinical onset, also inhibited established disease [23]. Because these doses did not significantly lower plasma cholesterol, the effect appeared largely independent of lipid lowering [23]. Findings with other statins were less favorable. In a later CIA study, atorvastatin and rosuvastatin had no effect on clinical, histologic, immunologic or biochemical markers of disease [24]. The benefit of i.p. simvastatin at 40 mg/kg/day, the dose used by Leung et al., was reproduced, but the authors attributed it to severe toxicity with raised glucocorticoid levels [24]. Deaths reduced this group from 18 to 7 mice over

the study, which suggests that the earlier benefit may reflect toxicity rather than a true antiarthritic effect [24]. When Vandebriel et al. gave CIA mice oral atorvastatin at 0.2 mg/day or oral pravastatin at 0.8 mg/day, arthritis started earlier, and every treated mouse became arthritic, against 7 of 12 controls [25]. After body-surface-area conversion for a 60-kg adult [7], the i.p. simvastatin doses of 10–40 mg/kg [23] correspond to approximately 49–195 mg/day, with the highest dose about five times the 40 mg/day used in RA trials [21, 22]. The oral atorvastatin dose used by Vandebriel et al., about 10 mg/kg, corresponds to approximately 49 mg/day and is close to the trial dose, whereas pravastatin at four times that dose corresponds to approximately 195 mg/day [25]. Earlier arthritis onset after statin treatment has, to our knowledge, been reported only by this group [25].

Results in patients are mixed. Two meta-analyses showed that statins reduced disease activity measured by DAS28, with standardized mean differences (SMDs) of −0.55 and −0.79 (95% CIs −0.83 to −0.26 and −1.08 to −0.49, respectively) [26, 27]. However, the included trials were small and mostly conducted at single centers [26], and both meta-analyses reported high heterogeneity

[26, 27]. The larger one also found possible publication bias for DAS28 (Egger test P = 0.012) [27]. In an unblinded randomized study conducted in Egypt, 30 patients with early RA were divided into two groups of 15 and treated for 6 months with MTX (0.2 mg/kg/week) and prednisone (10 mg/day), either alone (group 1) or with atorvastatin 40 mg/day (group 2). DAS28 was significantly reduced from 6.09 to 3.9 in group 2 compared with 6.19 to 5.27 in group 1 [28]. The double-blind Trial of Atorvastatin in Rheumatoid Arthritis (TARA) randomized 116 patients to atorvastatin 40 mg/day or placebo added to their existing DMARDs [21], with co-primary outcomes of change in DAS28 and European League Against Rheumatism (EULAR) response [21]. After 6 months, DAS28 improved more with atorvastatin than with placebo (between-group difference −0.52, P = 0.004; 95% CI from −0.87 to −0.17) [21]. This difference was slightly below the 0.6-unit change the trial was powered to detect, but EULAR response was more frequent with atorvastatin (31% vs 10%) [21], whereas the small unblinded Egyptian study reported a between-group difference of about 1.3 units [28]. In a 12-week randomized study without placebo

(blinding not reported), 55 patients with RA received atorvastatin 20 mg/day or no statin [29]. DAS28 fell by 0.41 with atorvastatin and by 0.03 without it ($P < 0.001$), and CRP and ESR also decreased [29]. The DAS28 change, however, was below the 0.6-unit threshold used to power TARA [21]. In contrast, the largest trial we found, the double-blind Trial of Atorvastatin for primary prevention of Cardiovascular Events in RA (TRACE RA), enrolled 3,002 UK patients for cardiovascular end points [22]. It found no difference in DAS28 between the atorvastatin and placebo groups [22]. Atorvastatin did lower CRP, but the authors judged the fall too small to matter clinically [22]. TRACE RA targeted cardiovascular end points, reported compliance in its atorvastatin arm fell from 89% at 3 months to 39% by 60 months, and the trial was stopped early because cardiovascular events were fewer than expected [22]. These design features, rather than a lack of efficacy alone, may partly explain why TRACE RA did not confirm the smaller trials.

Observational data on statin use and RA onset are inconsistent: in an Israeli cohort of 211,627 new statin users, persistence was measured using the proportion of days covered (PDC) [30]. Compared

with a PDC of <20%, a PDC of ≥80% was associated with lower RA risk (adjusted HR 0.58, 95% CI from 0.52 to 0.65), and to test for healthy-user bias, the same study also examined osteoarthritis [30]. For osteoarthritis, high persistence was associated with only a modest risk reduction (HR 0.85, 95% CI from 0.81 to 0.88) [30]. A US case–control study that matched 16,363 RA cases with the same number of controls initially found a slightly higher risk with statin use [31], but after adjustment for hyperlipidemia the association disappeared, with OR 0.95 (95% CI from 0.90 to 1.01) [31]. However, hyperlipidemia is often diagnosed when statin treatment begins, which may limit the accuracy of this adjustment [31]. Genetic data, in contrast, suggest a protective role. In a drug-target Mendelian randomization study, genetically proxied HMG-CoA reductase inhibition was linked to lower RA risk in both the discovery (OR 0.45, 95% CI from 0.32 to 0.63) and replication (OR 0.37, 95% CI from 0.23 to 0.61) datasets [32]. Although consistent, these data come from European populations only and address RA onset rather than disease activity in established RA [32]. A small prevention trial was also inconclusive: among 62 anti-citrullinated protein

antibody (ACPA)-positive patients with arthralgia, atorvastatin 40 mg/day was compared with placebo for a planned 3 years, but the trial was stopped early because of low inclusion [33]. Arthritis developed in 9/31 (29%) atorvastatin recipients and 6/31 (19%) placebo recipients (HR 1.40, 95% CI from 0.50 to 3.95), a non-significant difference [33]. Only 62 of the planned 220 participants were enrolled, which left the trial too small to confirm or exclude a protective effect [33]. For now, statins in RA are prescribed to lower lipids and cardiovascular risk, as in other patients [22], and larger, longer placebo-controlled trials with disease activity as the primary end point are needed before statins can be used to treat RA itself.

4.2.2. Fenofibrate

Fenofibrate is a hypolipidemic drug and a synthetic PPAR-α ligand [34]. In mouse T cells, fenofibrate (5–20 μM) inhibited Th17-cell differentiation in a dose-dependent manner, by 74% at 20 μM, whereas other PPAR-α agonists (WY14643, GW7647, and bezafibrate) had no such effect, which suggests that fenofibrate acts here through another pathway [35]. In AIA rats, fenofibrate (300 mg/kg/day for 12 days) lowered

the arthritis score from 11.0 to 5.83, a fall of about 47% ($P < 0.01$) [36]. This dose was far above clinical doses. By the body-surface-area method for a 60-kg adult, 300 mg/kg/day corresponds to approximately 2,900 mg/day, about 15–20 times the 145–200 mg/day used in RA studies [7, 36, 37, 38]. In a randomized study without a placebo arm, 44 patients with RA and dyslipidemia were assigned to fenofibrate 200 mg/day (n = 23) or statins (n = 21) for 6 months [38]. At 6 months, the prednisolone dose was lower in the fenofibrate arm than in the statin arm ($P = 0.0449$) [38]. Within the fenofibrate arm, it had fallen ($P = 0.0098$ vs baseline) from a mean of 3.6 (standard deviation [SD] 2.7) to 2.0 (SD 2.2) mg/day [38]. However, this small study tested several outcomes, so the result is exploratory. CRP changes were non-significant in both arms [38]. In an uncontrolled pilot study, 27 patients with active RA received 145 mg/day of fenofibrate for 12 weeks in addition to stable DMARDs, and median DAS28 fell from 6.28 to 4.83 [37]. Without a control group, part of this fall may reflect regression to the mean in patients enrolled during high disease activity. CRP and IL-6 decreased by 36% and 41%, respectively, while 2 patients

withdrew because of severe back pain considered possibly drug-related [37]. Clinical evidence therefore remains limited to these two small studies, and to our knowledge no placebo-controlled trial has evaluated fenofibrate in RA.

References

1. Chung CP, Oeser A, Solus JF, et al. Inflammation-Associated Insulin Resistance: Differential Effects in Rheumatoid Arthritis and Systemic Lupus Erythematosus Define Potential Mechanisms. Arthritis and rheumatism. 2008. doi:10.1002/art.23600

2. Zhou G, Myers R, Li Y, et al. Role of AMP-activated protein kinase in mechanism of metformin action. The Journal of clinical investigation. 2001. doi:10.1172/JCI13505

3. Gwinn D, Shackelford D, Egan D, et al. AMPK phosphorylation of raptor mediates a metabolic checkpoint. Molecular cell. 2008. doi:10.1016/j.molcel.2008.03.003

4. Nagai S, Kurebayashi Y, Koyasu S. Role of PI3K/Akt and mTOR complexes in Th17 cell differentiation. Annals of the New York Academy of Sciences. 2013. doi:10.1111/nyas.12059

5. Son HJ, Lee J, Lee SY, et al. Metformin

Attenuates Experimental Autoimmune Arthritis through Reciprocal Regulation of Th17/Treg Balance and Osteoclastogenesis. Mediators of Inflammation. 2014. doi:10.1155/2014/973986
6. Fan KJ, Wu J, Wang QS, et al. Metformin inhibits inflammation and bone destruction in collagen-induced arthritis in rats. Annals of Translational Medicine. 2020. doi:10.21037/atm-20-3042
7. Nair AB, Jacob S. A simple practice guide for dose conversion between animals and human. Journal of Basic and Clinical Pharmacy. 2016. doi:10.4103/0976-0105.177703
8. Naffaa M, Rosenberg V, Watad A, et al. Adherence to metformin and the onset of rheumatoid arthritis: a population-based cohort study. Scandinavian Journal of Rheumatology. 2020. doi:10.1080/03009742.2019.1695928
9. Chen Y, Qiu F, Yu B, et al. Metformin, an AMPK Activator, Inhibits Activation of FLSs but Promotes HAPLN1 Secretion. Molecular Therapy. Methods & Clinical Development. 2020. doi:10.1016/j.omtm.2020.05.008
10. Abdallah MS, Alarfaj SJ, Saif DS, et al. The AMPK modulator metformin as adjunct to methotrexate in patients with rheumatoid arthritis:

A proof-of-concept, randomized, double-blind, placebo-controlled trial. International Immunopharmacology. 2021. doi:10.1016/j.intimp.2021.107575
11. Gharib M, Elbaz W, Darweesh E, et al. Efficacy and Safety of Metformin Use in Rheumatoid Arthritis: A Randomized Controlled Study. Frontiers in Pharmacology. 2021. doi:10.3389/fphar.2021.726490
12. Zhang L, Zhou Y, Jiang S, et al. Effects of metformin therapy on HMGB1 levels in rheumatoid arthritis patients. European Journal of Medical Research. 2023. doi:10.1186/s40001-023-01476-x
13. Alshammari MM, Abood NA, Kahwaji S, et al. Effectiveness of metformin as an adjunctive therapy in rheumatoid arthritis: a systematic review and meta-analysis of randomized controlled trials. BMC Rheumatology. 2026. doi:10.1186/s41927-026-00659-3
14. Zemedikun DT, Gokhale K, Chandan JS, et al. Type 2 diabetes mellitus, glycaemic control, associated therapies and risk of rheumatoid arthritis: a retrospective cohort study. Rheumatology. 2021. doi:10.1093/rheumatology/keab148

15. Liang J, Cai Y, Zhang J, et al. Metformin Treatment Reduces the Incidence of Rheumatoid Arthritis: A Two-Sample Mendelian Randomized Study. Journal of Clinical Medicine. 2023. doi:10.3390/jcm12072461
16. Hansildaar R, Vedder D, Baniaamam M, et al. Cardiovascular risk in inflammatory arthritis: rheumatoid arthritis and gout. The Lancet. Rheumatology. 2021. doi:10.1016/S2665-9913(20)30221-6
17. Greenwood J, Steinman L, Zamvil S. Statin therapy and autoimmune disease: from protein prenylation to immunomodulation. Nature reviews. Immunology. 2006. doi:10.1038/nri1839
18. Kwak B, Mulhaupt F, Myit S, et al. Statins as a newly recognized type of immunomodulator. Nature medicine. 2000. doi:10.1038/82219
19. Gao Y, Zhang Y, Liu X. Rheumatoid arthritis: pathogenesis and therapeutic advances. MedComm. 2024. doi:10.1002/mco2.509
20. Walker ME, Souza PR, Colas RA, et al. 13-Series resolvins mediate the leukocyte-platelet actions of atorvastatin and pravastatin in inflammatory arthritis. The FASEB Journal. 2017. doi:10.1096/fj.201700268
21. McCarey D, McInnes I, Madhok R, et al. Trial

of Atorvastatin in Rheumatoid Arthritis (TARA): double-blind, randomised placebo-controlled trial. The Lancet. 2004. doi:10.1016/S0140-6736(04)16449-0
22. Kitas GD, Nightingale P, Armitage J, et al. A Multicenter, Randomized, Placebo-Controlled Trial of Atorvastatin for the Primary Prevention of Cardiovascular Events in Patients With Rheumatoid Arthritis. Arthritis & Rheumatology. 2019. doi:10.1002/art.40892
23. Leung B, Sattar N, Crilly A, et al. A novel anti-inflammatory role for simvastatin in inflammatory arthritis. Journal of immunology. 2003. doi:10.4049/jimmunol.170.3.1524
24. Palmer G, Chobaz V, Talabot-Ayer D, et al. Assessment of the efficacy of different statins in murine collagen-induced arthritis. Arthritis and rheumatism. 2004. doi:10.1002/art.20673
25. Vandebriel RJ, De Jong HJ, Gremmer ER, et al. Statins accelerate the onset of collagen type II-induced arthritis in mice. Arthritis Research & Therapy. 2012. doi:10.1186/ar3814
26. Xing B, Yin YF, Zhao LD, et al. Effect of 3-Hydroxy-3-Methylglutaryl-Coenzyme A Reductase Inhibitor on Disease Activity in Patients With Rheumatoid Arthritis. Medicine. 2015.

doi:10.1097/MD.0000000000000572
27. Ren C, Li M. The efficacy of statins in the treatment of rheumatoid arthritis: A systematic review and meta-analysis. Medicine. 2023. doi:10.1097/MD.0000000000035088
28. El-Barbary AM, Hussein MS, Rageh EM, et al. Effect of Atorvastatin on Inflammation and Modification of Vascular Risk Factors in Rheumatoid Arthritis. The Journal of Rheumatology. 2011. doi:10.3899/jrheum.100582
29. Tang TT, Song Y, Ding YJ, et al. Atorvastatin upregulates regulatory T cells and reduces clinical disease activity in patients with rheumatoid arthritis. Journal of Lipid Research. 2011. doi:10.1194/jlr.M010876
30. Chodick G, Amital H, Shalem Y, et al. Persistence with Statins and Onset of Rheumatoid Arthritis: A Population-Based Cohort Study. PLoS Medicine. 2010. doi:10.1371/journal.pmed.1000336
31. Peterson MN, Dykhoff HJ, Crowson CS, et al. Risk of rheumatoid arthritis diagnosis in statin users in a large nationwide US study. Arthritis Research & Therapy. 2021. doi:10.1186/s13075-021-02617-5
32. Hu X, Zhang P, Gao Y, et al. Identification of

lipid-modifying drug targets for autoimmune diseases: insights from drug target mendelian randomization. Lipids in Health and Disease. 2024. doi:10.1186/s12944-024-02181-2
33. van Boheemen L, Turk S, van Beers-Tas M, et al. Atorvastatin is unlikely to prevent rheumatoid arthritis in high risk individuals: results from the prematurely stopped STAtins to Prevent Rheumatoid Arthritis (STAPRA) trial. RMD Open. 2021. doi:10.1136/rmdopen-2021-001591
34. Varga T, Czimmerer Z, Nagy L. PPARs are a unique set of fatty acid regulated transcription factors controlling both lipid metabolism and inflammation. Biochimica et Biophysica Acta. 2011. doi:10.1016/j.bbadis.2011.02.014
35. Zhou Z, Sun W, Liang Y, et al. Fenofibrate Inhibited the Differentiation of T Helper 17 Cells In Vitro. PPAR Research. 2012. doi:10.1155/2012/145654
36. Castillero E, Martín AI, Nieto-Bona MP, et al. Fenofibrate administration to arthritic rats increases adiponectin and leptin and prevents oxidative muscle wasting. Endocrine Connections. 2012. doi:10.1530/EC-12-0003
37. Shirinsky I, Polovnikova O, Kalinovskaya N, et al. The effects of fenofibrate on inflammation

and cardiovascular markers in patients with active rheumatoid arthritis: a pilot study. Rheumatology International. 2013.
doi:10.1007/s00296-012-2613-z
38. Goto M. A comparative study of anti-inflammatory and antidyslipidemic effects of fenofibrate and statins on rheumatoid arthritis. Modern Rheumatology. 2010.
doi:10.1007/s10165-009-0261-2